Zir Image Enhancer: phone photos to studio packshots

Team D3A24 · Sartiq challenge · Neural Wave 2026

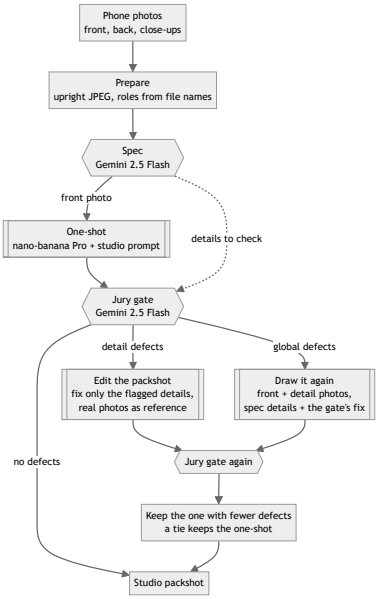
1. What we built

A pipeline that turns one to five phone photos of a garment (on a hanger, a mannequin, worn or on the floor) into one front-view studio packshot, and checks it for the defects a buyer would notice before accepting it. It runs as a Gradio app (upload, before/after, a log of what it understood and checked; photo roles come from the file names) and as a batch command, `zir-run`, that writes the submission layout.

Model (all through fal.ai)	Role
Gemini 2.5 Flash (vision)	Reads all photos once (the *spec*): which photo is the front, garment type, details to keep (logos, text, buttons), proportions, how the photo was taken, pieces of a set. Later checks the result (the *gate*).
nano-banana Pro edit	Draws the packshot from the front photo and a written studio prompt (nano-banana 2.1 is the cheaper option).

2. The final pipeline (v3)

1. **Prepare** the photos: HEIC, EXIF rotation and transparency fixed; roles read from file names (`03_detail_chest-logo.jpg`).
2. **Spec**: one Gemini call with every photo.
3. **One-shot**: our studio prompt and the front photo only. Mannequin photos get an invisible-mannequin variant of the prompt.
4. **Jury gate**: Gemini compares the packshot with the real photo on the criteria the Sartiq jury named (crop, logos and text, zips and buttons, seams, texture, colour), plus same garment, front view, proportions, studio light, extra items, missing pieces and invented text.
5. **Repair only when needed**, matched to the defect: a *detail* defect is fixed by **editing the packshot** with the real photos as reference, so shape, colour and framing stay; a *global* defect (colour, proportions, view, crop) by **drawing it again** with the spec's details and the gate's fix.
6. **Keep the version with fewer defects**; severe ones count double, and a tie keeps the one-shot.



About 20 seconds and $0.16 per garment with Pro ($0.06 with 2.1); a repair is needed for about one garment in ten. Every paid call is cached by content, so nothing is paid twice, and a hard budget limit stops spending.

3. How we got here

Version	What changed	Holdout (38)	Dev	Per garment
Phone photo unchanged		39.78	48.37	
Official baseline prompt		60.22	68.71	$0.06
v1	Our studio prompt (written from the training targets), portrait canvas, Gemini picks the front	61.82	68.62	$0.06
v2	Spec details, proportions and sets in the prompt, named detail photos, gate, one redraw	61.77	68.30	$0.07
v2-branch: LoRA	Fine-tuned FLUX.2 klein 9B (step-1750, rank 128) on v2 harness, 327 curated pairs	51.01	not submitted	$0.10
v3	v1 first, then the jury gate; edit for detail defects, redraw for global ones	61.90	68.75	$0.06
v4	9 versions per garment (prompts × seeds), one picked by CLIP + AlexNet embeddings, gate on the top 3	63.01	69.59	$0.55
v3 + Pro (final)	v3 with nano-banana Pro	not run	**69.77**	$0.16

v2 put everything the spec knew into the prompt. It scored slightly lower: the edit model does best with a short instruction, and redrawing for small defects moved shape and colour. **v3** kept the short one-shot and only repaired what the jury would see, choosing an edit or a redraw by defect type; on the holdout it fixed a wrong colour (+10.4), an embroidered logo (+0.8) and a texture (+0.9).

v4 came from noticing that nano-banana gives a different image every call: on the holdout the best of nine would score 65.9. To pick without the target we embedded every candidate with CLIP (what the garment is, overall style) and AlexNet (the network inside LPIPS, so it sees texture as the scorer does), and ranked by closeness to the ten nearest Sartiq training photos (the house style) and to the garment's own photo. A Gemini judge, the scorer run against the input photo and a consensus vote had all picked no better than random, because none of them knows Sartiq's style. v4 gained about +0.2 per distinct version, but at nine images per garment.

Parallel track: Fine-tuning after v2. After v2, we branched to explore whether domain fine-tuning could beat commercial prompt orchestration. We trained a LoRA on FLUX.2 klein 9B across 327 curated training pairs on vast.ai (filtering 61 noisy/colour-drifted pairs and strictly isolating our 38 holdout pairs). Evaluated on the v2-era harness (without the later v3 jury gate and hybrid repair), the adapter improved substantially over the base FLUX.2 model (51.01 vs 43.36), proving the model learned packshot geometry. However, it still lagged behind our hosted baseline, suffering most on colour fidelity (0.477 vs 0.620). Furthermore, FLUX 9B carries a non-commercial licence. Seeing this gap, we stopped the training track and focused our remaining time on the main branch, developing the v3 jury gate and moving to nano-banana Pro.

v3 + Pro is where we ended. In v4's pool, Pro versions lowered dev (69.34 and 68.89 against 69.59), but run as v3 with the same settings it scored 69.77 against 68.64 with 2.1, above v4 at less than a third of its cost. Its first image also had gate defects on 2 of 27 dev garments against 4. Our submission for the 15 on-site garments is v3 + Pro: $2.71 in total, every image checked by hand with no visible defect; the organisers score it against their private references.

4. How we evaluated

The automatic score compares the output with a hidden studio photo (colour 40%, LPIPS 25%, CLIP 15%, silhouette IoU 15%, SSIM 5%). We carved a 38-pair holdout out of the 382 training pairs, grouped by target so no studio photo appears on both sides, and scored every idea with the official scorer before spending on the next one; dev scores come from the leaderboard. Dev is 67% mannequin photos, test is all hanger photos and our holdout 26% mannequin, so some holdout gains did not move dev, and the other way round.

5. What we tried and dropped

Every result is in `experiments/results.csv`; every prompt with its scores in `docs/PROMPTS.md`.

Idea	Result	Why
FLUX.2 klein 9B + LoRA (v2 branch)	51.01 (+7.65 over base FLUX.2)	Learned flat-lay geometry, but colour drifted (-0.143 vs harness); some garments have heavy generation artifacts
Zero-shot generators: FLUX.2, Seedream 4.5, Qwen-Image-Edit (with and without an outfit LoRA)	43.4, 49.6, 54.4, 39.7	They change the garment
Pre-processing: cut-out, crop, denoise, 2x super-resolution	-3.9, -1.3, -1.9, -3.0	The edit model reads the whole scene; any change to the photo hurts
Colour and light fixes after generation; a per-garment colour predictor	-0.1 to -14; predictor no better than none	The studio colour cannot be recovered from the phone photo
Longer prompts: spec-written, flatten mannequin shapes, real length, vivid colour, relight pass	-3.0, -1.2, -1.3, -1.5	Short instructions work best
Prompt wording variants on dev	±0.1	Choosing or changing the model moved more than wording
DINOv2 as a third expert; embeddings before the generator	no gain	Too few scored garments to learn per-garment choices

6. Capture protocol

The front photo first, one of the back, and a close-up of every logo, print, label or closure, named in the file name (`03_detail_chest-logo.jpg`); garment on a plain surface with soft, even light. Named close-ups are the only extra photos the generator sees, because unnamed ones pulled colour (-2.1). On 8 garments of our own shot on purpose from messy to tidy, the spec picked the tidy photo every time and all 8 came out clean.

7. Limitations and next steps

- Colour under strong coloured light stays a guess (a grey card in the frame would remove it).
- The gate is a vision model, not a measurement: strict on some seams, and it sometimes asks for back details on the front; the rule "a tie keeps the one-shot" keeps the safer image.
- Several versions plus the embedding selector (v4) is the next step if cost allows: with Pro it would cost about $1.40 per garment.
- Past the generator's plateau: a model of our own (a frozen DINOv2/SigLIP encoder per photo, aggregated across views, conditioning a diffusion decoder with a silhouette control, trained with a colour term aligned with the scorer), which needs data and GPU time we did not have.
- Fine-tuning potential: Our LoRA was a single exploratory run tested only on the earlier v2 harness. Despite these constraints, gaining +7.65 points over the base FLUX.2 model (43.36 → 51.01) shows that custom domain adaptation is a viable path. With more time to tune training parameters, evaluate across checkpoints, and integrate the adapter into our mature v3 gating and repair pipeline, fine-tuning remains a promising long-term direction.